Departement Elektriese, Elektroniese en Rekenaar-Ingenieurswese

Department of Electrical, Electronic and Computer Engineering

Semestertoets 3
VRAESTEL

Semester Test 3
QUESTION PAPER

Module EBN122
Elektrisiteit en Elektronika
Oktober 2017

Module EBN122
Electricity and Electronics
October 2017

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Memo.

Eksamensinligting: *Exam information:*			
Maksimum punte: *Maximum marks:*	46	Volpunte: *Full marks:*	45
Duur van vraestel: *Duration of paper:*	90 min + 10 min to complete OCR form *90 min + 10 min om OKH vorm te voltooi*	Oopboek / toeboek: *Open / closed book:*	Toe *Closed*
Totale aantal bladsye (hierdie blad ingesluit): *Total number of pages (including this page):*	19 + OCR		

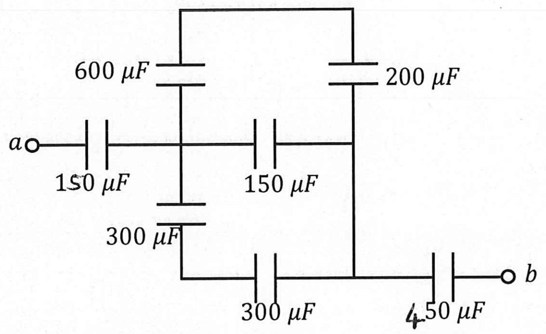
BELANGRIK- ***IMPORTANT***

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
   The examination regulations of the University of Pretoria apply.
2. **Alle vrae moet op die ANTWOORDBLAD en op die Optiese Karakter Herkenning (OKH) vorm ingevul word.**
   All questions must be answered on the ANSWER SHEET and on the Optical Character Recognition (OCR) form.
3. Neem noukeurig kennis van die instruksies op die OKH vorm.
   Take careful consideration of the instructions on the OCR form.

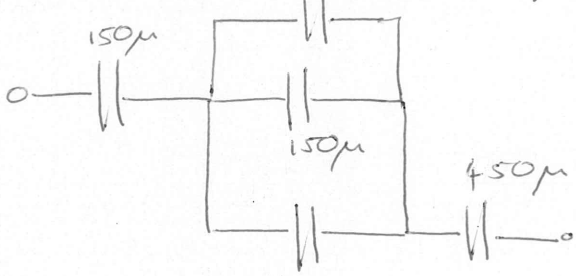
Dosent(e): *Lecturer(s):*	1. D le Roux
	2. D Ramotsoela
	3. X Ye

SECTION / AFDELING 1 [10]

Question 1 [2]		Vraag 1 [2]	
01	Calculate C_{ab}.	**01**	Bereken C_{ab}.

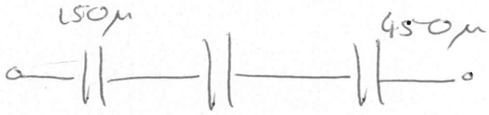


$600\mu \parallel 200\mu = 150\mu$



$300 \parallel 300$

$= 150\mu$



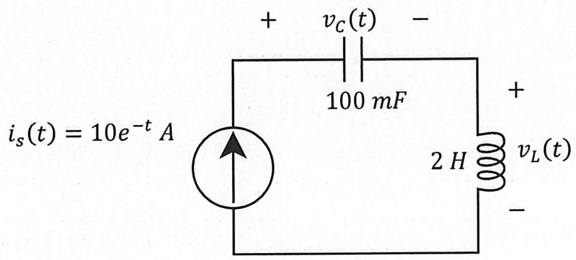
$150 + 150 + 150$

$= 450\mu$

$C_{ab} = 150\mu \parallel 450\mu \parallel 450\mu$

$= 90\mu F.$

Question 2 & 3 [4]		Vraag 2 & 3 [4]	
For the circuit below, calculate the voltage across the capacitor and the voltage across the inductor when $t = 0.1$ s. Assume $v_L(0) = -20$ V, and $v_C(0) = 0$ V.		Vir onderstaande kringbaan, bereken die spanning oor die kapasitor en die spanning oor die induktor vir $t = 0.1$ s. Aanvaar $v_L(0) = -20$ V, en $v_C(0) = 0$ V.	
02	v_C (0.1s)	**02**	v_C (0.1s)
03	v_L (0.1s)	**03**	v_L (0.1s)



$$v_L = L\frac{di}{dt}$$

$$= 2\frac{d}{dt}(10e^{-t})$$

$$= -20e^{-t}$$

$$v_L(0.1) = -20e^{-0.1}$$

$$= \underline{-18.097\ V}$$

$$v_C = \frac{1}{C}\int_0^t i\,dt + v_C(0)$$

$$= \frac{1}{0.1}\int_0^t 10e^{-t}\,dt$$

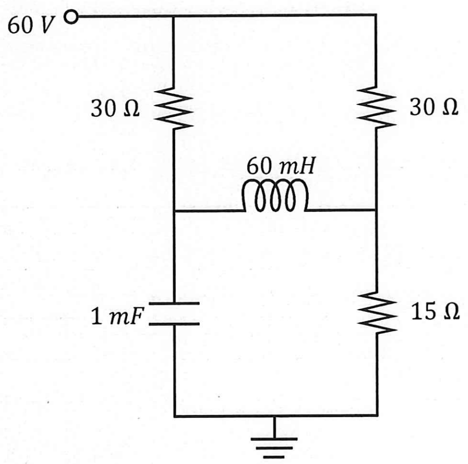
$$= 100\int_0^t e^{-t}\,dt$$

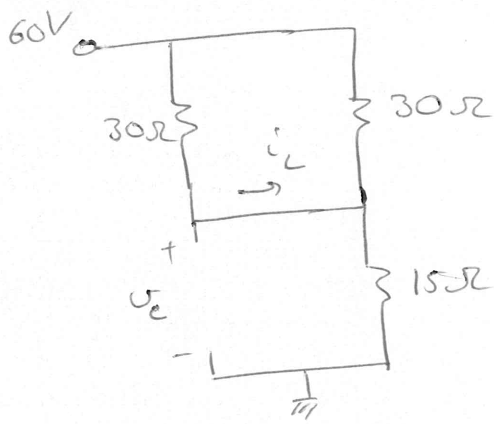
$$= 100\left[-e^{-t}\right]_0^t$$

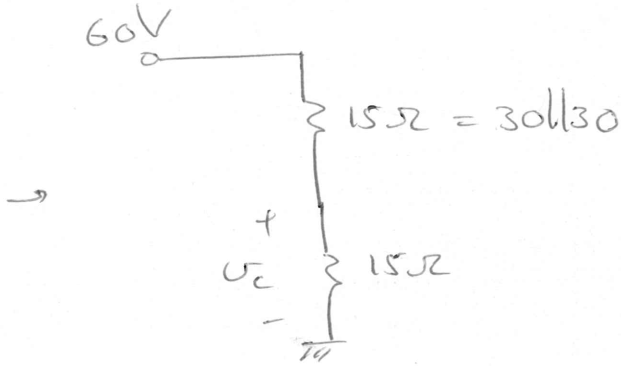
$$= 100(1-e^{-t})$$

$$v_C(0.1) = \underline{9.516\ V}$$

Question 4 & 5 [4]		Vraag 4 & 5 [4]	
Assume steady-state conditions (Direct Current) for the circuit below. What is the energy stored by the capacitor w_C and inductor w_L?		Aanvaar gestadigde toestand (Gelykstroom) vir onderstaande kringbaan. Wat is die energie gestoor deur die kapasitor w_C en induktor w_L?	
04	w_C	**04**	w_C
05	w_L	**05**	w_L







* The 60V splits equally over the two 15Ω resistors.

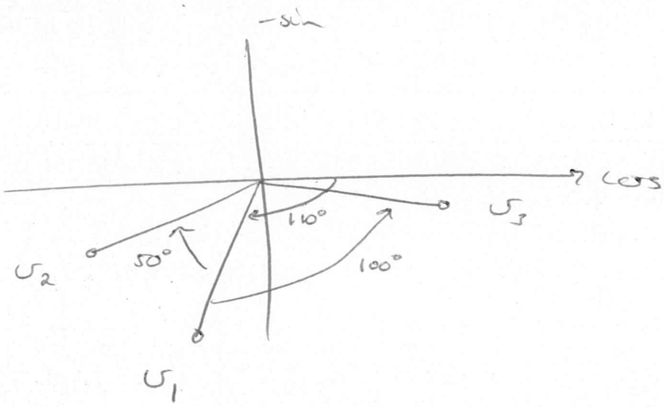
$\therefore$ $v_c = 30V$ | $i_L = \frac{30V}{30\Omega} = 1A$.

$w_c = \frac{1}{2}(1m)(30)^2 = \underline{450mJ}$

$w_L = \frac{1}{2}(60m)(1)^2 = \underline{30mJ}$

SECTION / AFDELING 2 [18]

Questions 6 & 7 [4]		Vraag 6 & 7 [4]	
The following sinusoid is given: $v_1 = 16\cos(\omega t - 110°)\ V$ It is known that: - v_1 leads v_2 by $50°$. - v_1 lags v_3 by $100°$.		Die volgende sinusied word gegee: $v_1 = 16\cos(\omega t - 110°)\ V$ Dit is bekend dat: - v_1 loop voor v_2 met $50°$. - v_1 loop agter v_3 met $100°$.	
06	Choose the correct statement: 1. v_2 leads v_3 2. v_3 leads v_2 (Write ***1*** if statement 1 is correct, and ***2*** if statement 2 is correct.)	**06**	Kies die korrekte stelling: 1. v_2 loop voor v_3 2. v_3 loop voor v_2 (Skryf ***1*** indien stelling 1 korrek is, en ***2*** indien stelling 2 korrek is.)
07	If $v_2 = 6\cos(wt + \theta)\ V$, what is θ? (Give the angle in the range of –180° to 180°.)	**07**	Indien $v_2 = 6\cos(wt + \theta)\ V$, wat is θ? (Gee die hoek in die bereik van –180° tot 180°.)



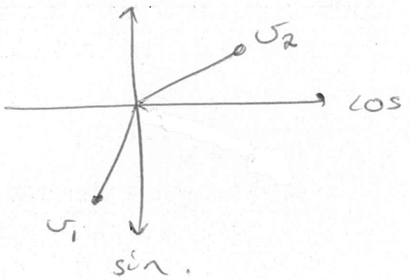
$\therefore v_3$ leads v_2.

$\theta = -160°$

Questions 8 & 9 [4]		Vraag 8 & 9 [4]	
08	Solve for z: $$z = \frac{\sqrt{-j4}}{(1.732 - j)^*}$$ (Write the final answer in **polar form.**)	**08**	Los op vir z: $$z = \frac{\sqrt{-j4}}{(1.732 - j)^*}$$ (Skryf die finale antwoord in **polêre vorm.**)
09	The following sinusoids are given: $$v_1 = 6\sin(\omega t - 20.05°)\ V$$ $$v_2 = 5.46\cos(\omega t + 40.07°)\ V$$ Determine the phasor value of $\bar{V}_T = \bar{V}_1 + \bar{V}_2$. Give the answer in **polar form.**	**09**	Die volgende sinusiede word gegee: $$v_1 = 6\sin(\omega t - 20.05°)\ V$$ $$v_2 = 5.46\cos(\omega t + 40.07°)\ V$$ Bepaal die fasor waarde van $\bar{V}_T = \bar{V}_1 + \bar{V}_2$. Gee die antwoord in **polêre vorm.**

$$z = \frac{\sqrt{4\angle -90^\circ}}{1.732 + j} = \frac{2\angle -45^\circ}{\sqrt{1.732^2 + 1^2}\ \angle \tan^{-1}\left(\frac{1}{1.732}\right)}$$

$$= \frac{2\angle -45^\circ}{2\angle 30^\circ} = \underline{1\angle -75^\circ}\ 2$$



$$v_1 = 6\cos(\omega t - 20.05^\circ - 90^\circ)$$

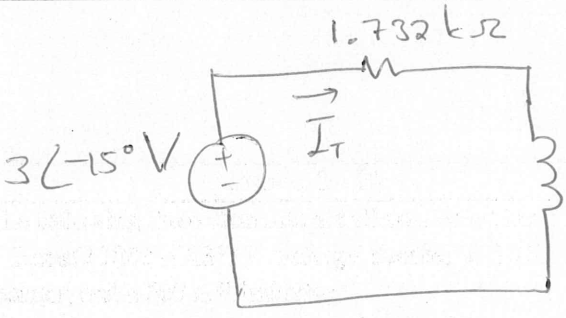
$$\therefore V_1 = 6\angle -110.05^\circ$$

$$v_2 = 5.46\cos(\omega t + 40.07^\circ)$$

$$\therefore V_2 = 5.46\angle +40.07^\circ$$

$$V_T = V_1 + V_2 = \underline{3\angle -45^\circ\ V}\ 2$$

Question 10 [2]		Vraag 10 [2]	
The following three elements are all connected in series: a 3 $\cos(2000t - 15°)\ V$ voltage source, a 1.732 kΩ resistor, and a 500 mH induktor. What is the total current through the circuit? (Give the answer in ~~**rectangular**~~ polar **form.**)		Die volgende drie elemente word in series gekoppel: 'n 3 $\cos(2000t - 15°)\ V$ spanningsbron, 'n 1.732 kΩ resistor, en 'n 500 mH induktor. Wat is die totale stroom deur die kringbaan? (Skryf die antwoord in ~~**reghoekige**~~ polar **vorm.**)	
10	$\bar{I}_T$ polar	**10**	$\bar{I}_T$ polar



$Z_L = j\omega L = j(2000)(0.5)$

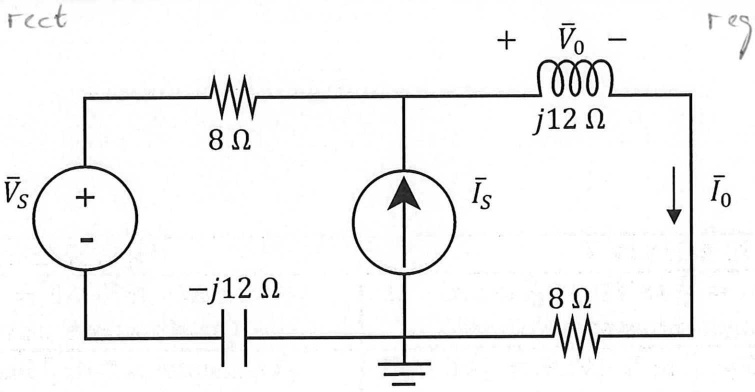
$= j1\ k\Omega$

$$\therefore I_T = \frac{3\angle -15°}{1.732k + j1k} = \frac{3\angle -15°}{(2\angle 30°)k}$$

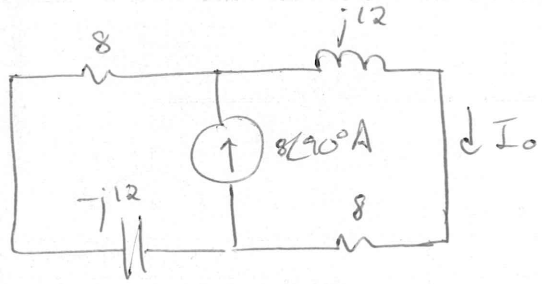
$$= 1.5\angle -45°\ mA$$

✓✓

Question 11 & 12 [4]		Vraag 11 & 12 [4]	
11	If $\bar{V}_S = 0\ V$ and $\bar{I}_S = 8\angle 90°\ A$, determine $\bar{I}_0$. (Write the answer in **rectangular form.**)	**11**	Indien $\bar{V}_S = 0\ V$ en $\bar{I}_S = 8\angle 90°\ A$, bepaal $\bar{I}_0$. (Skryf die antwoord in **reghoekige vorm.**)
12	If $\bar{V}_S = 8\angle 90°\ V$ and $\bar{I}_S = 0\ A$, determine $\bar{V}_0$. (Write the answer in ~~**polar form.**~~) rect	**12**	If $\bar{V}_S = 8\angle 90°\ V$ en $\bar{I}_S = 0\ A$, bepaal $\bar{V}_0$. (Skryf die antwoord in ~~**polêre vorm.**~~) regh.



11



* I-division:

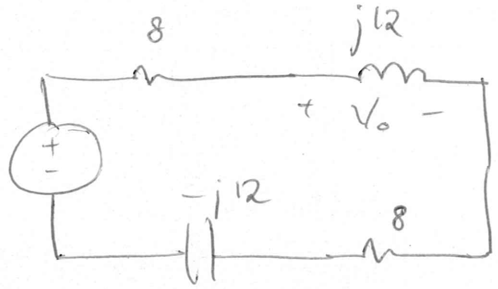
$$I_o = \frac{8\angle 90° (8 - j12)}{(8 - j12) + (8 + j12)}$$

$$= \frac{8j (8 - j12)}{16}$$

$$= 6 + j4 \ A$$

12

8∠90°V



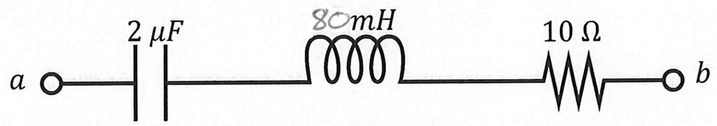
* V-division:

$$V_o = \frac{8\angle 90° \times j12}{8 + 8 + j12 - j12}$$

$$= \frac{8j \times j12}{16}$$

$$= -6 \ V.$$

Question 13 [2]		Vraag 13 [2]	
13	For wat value of ω will Z_{ab} only be real? (I.e. $Z_{ab} = R + j0\ \Omega$)	13	Vir watter waarde van ω sal Z_{ab} slegs reëel wees? (M.a.w. $Z_{ab} = R + j0\ \Omega$)



$$Z_{ab} = Z_C + Z_L + Z_R$$

$$= \frac{-j}{\omega C} + j\omega L + R$$

But Z_{ab} must only be equal to R.

$$\therefore \frac{-j}{\omega C} + j\omega L = 0.$$

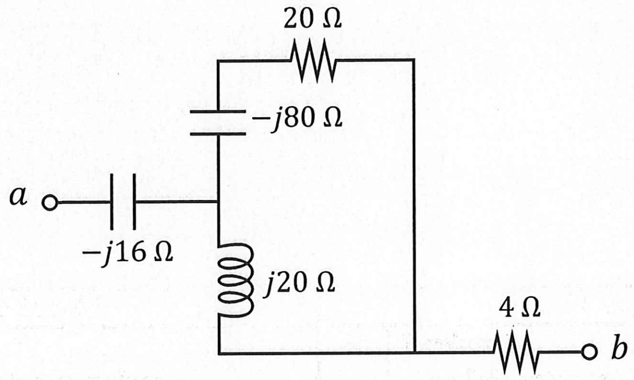
$$\omega L = \frac{1}{\omega C}.$$

$$\omega^2 = \frac{1}{L \times C}$$

$$\omega = \frac{1}{+\sqrt{80m \times 2\mu}}$$

$$= 2.5 \text{ krad/s}.$$

Question 14 [2]		Vraag 14 [2]	
14	Calculate Z_{ab}. (Give the answer in **rectangular form**.)	14	Bereken Z_{ab}. (Skryf die antwoord in **reghoekige vorm**.)



$$Z_{ab} = -j16 + (j20) \| (-j80 + 20) + 4$$

$$= 4 - j16 + \frac{(j20)(20 - j80)}{20 - j60}$$

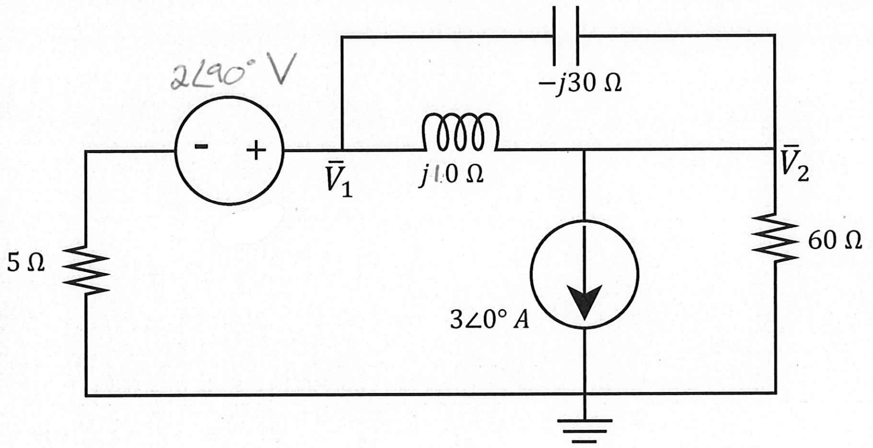
$$= 4 - j16 + 2 + 26j$$

$$= \underline{6 + 10j \ \Omega}$$

SECTION / AFDELING 3 [18]

Question 15-18 [4]	Vraag 15-18 [4]
Use nodal analysis to set up two equations of the following form $a\bar{V}_1 + b\bar{V}_2 = j12$ $c\bar{V}_1 + d\bar{V}_2 = -180$ Give the coefficients in **rectangular form**.	Gebruik node analise om twee vergelykings van die volgende vorm op te stel: $a\bar{V}_1 + b\bar{V}_2 = j12$ $c\bar{V}_1 + d\bar{V}_2 = -180$ Gee die koëffisiënte in **reghoekige vorm**.

15	*a*	**16**	*b*	**15**	*a*	**16**	*b*
17	*c*	**18**	*d*	**17**	*c*	**18**	*d*



① $\dfrac{V_1 - 2\angle 90^\circ}{5} + \dfrac{V_1 - V_2}{j10} + \dfrac{V_1 - V_2}{-j30} = 0.$

$(\times 30)\quad 6V_1 - j12 - j6V_1 + j6V_2 + jV_1 - jV_2 = 0.$

$V_1(6 - j5) + V_2(j5) = j12.$

② $3 + \dfrac{V_2}{60} + \dfrac{V_2 - V_1}{j10} + \dfrac{V_2 - V_1}{-j30} = 0$

$(\times 60)\quad V_2 - j6V_2 + j6V_1 + j2V_2 - j2V_1 = -180$

$(+j4)V_1 + (1 - j4)V_2 = -180$

(Remember: $\frac{1}{j} = -j$ and $\frac{1}{-j} = j$.)

Question 19-22 [4]

Use mesh analysis to set up two equations of the following form:

$$e\bar{I}_1 + f\bar{I}_2 = 16 - j12$$
$$g\bar{I}_1 + h\bar{I}_2 = 2\ A$$

Give the coefficients in **rectangular form**.

19	e	**20**	f
21	g	**22**	h

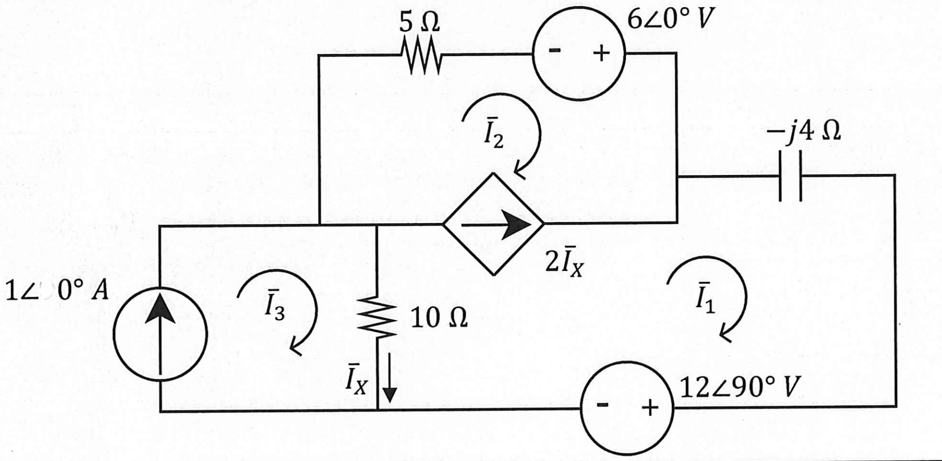
Vraag 19-22 [4]

Gebruik lusanalise om twee vergelykings in die volgende vorm op te stel:

$$e\bar{I}_1 + f\bar{I}_2 = 16 - j12$$
$$g\bar{I}_1 + h\bar{I}_2 = 2\ A$$

Gee die koëffisiënte in **reghoekige vorm**.

19	e	**20**	f
21	g	**22**	h



* $I_3 = 1A$. $\quad I_x = I_3 - I_1$

* Supermesh: $5I_2 - 6 + (-j4)I_1 + j12$
$+10I_1 - 10I_3 = 0$

$(10-j4)I_1 + 5I_2 = 6 - j12 + 10(1)$

$\underline{(10-j4)}I_1 + \underline{5}I_2 = 16 - j12$

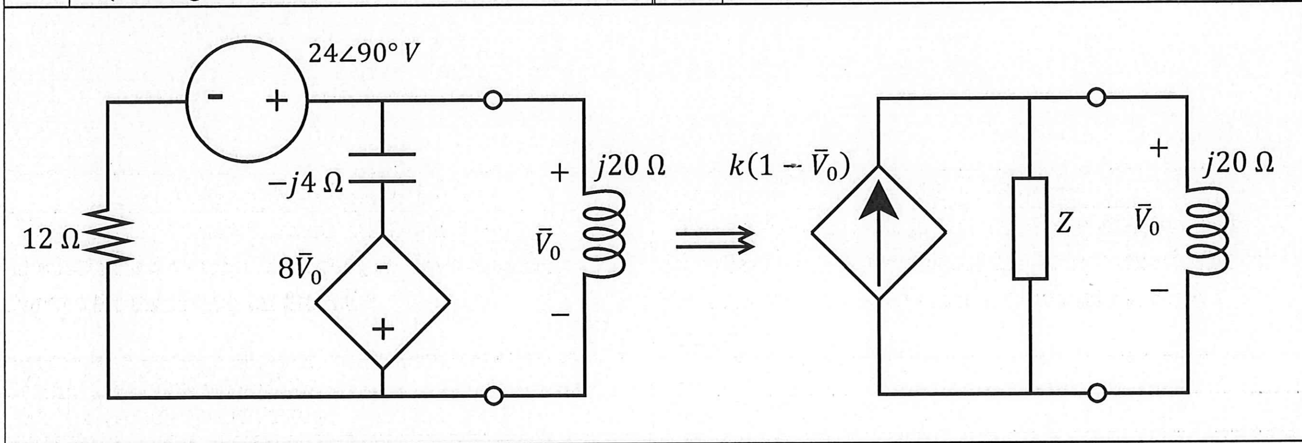
* KCL: $2I_x = I_1 - I_2$

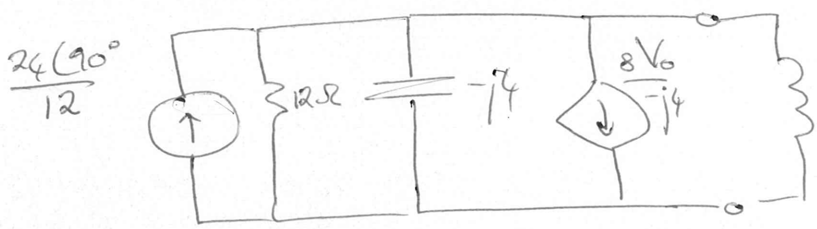
$2(I_3 - I_1) = I_1 - I_2$

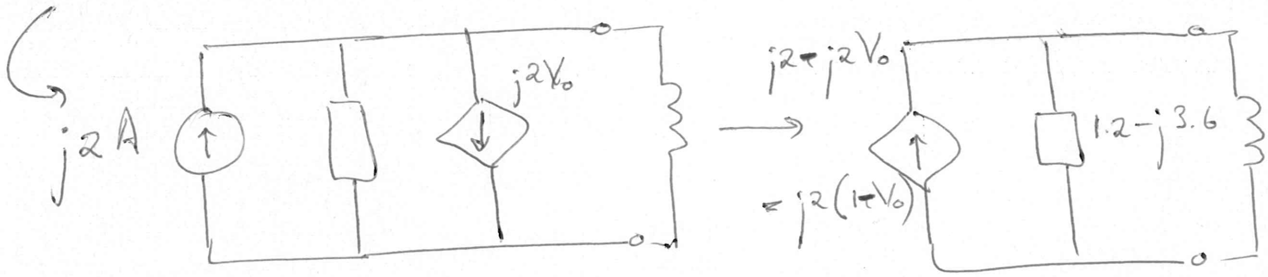
$2 = I_1 + 2I_1 - I_2$

$2 = \underline{3}I_1 \underline{-}I_2$

Question 23 & 24 [2]		Vraag 23 & 24 [2]	
Using source transformation, transform the circuit on the left into the circuit on the right. Give the unknown values for the circuit on the right.		Transformeer die kringbaan aan die linkerkant na die kringbaan aan die regterkant d.m.v. brontransformasie. Gee die onbekende waardes vir die regterkantste kringbaan.	
23	**k (rectangular form)**	**23**	**k (reghoekige vorm)**
24	**Z (rectangular form)**	**24**	**Z (reghoekige vorm)**







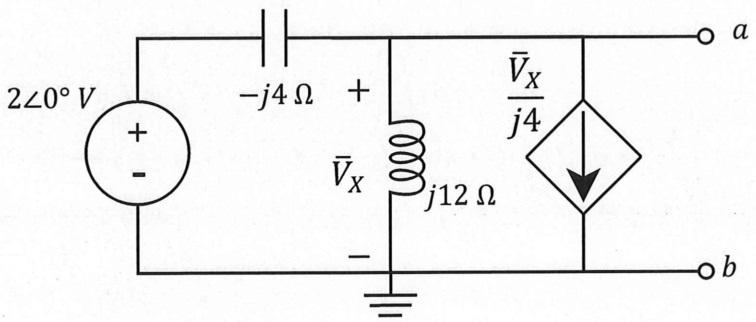
$\therefore k = j2$

$$Z_{11} = 12 \parallel -j4$$

$$= \frac{12(-j4)}{12 - j4}$$

$$= \underline{1.2 - j3.6\ \Omega.}$$

Question 25 & 26 [4]		Vraag 25 & 26 [4]	
Calculate the Thevenin equivalent with respect to terminals *a*-*b*. rect		Bereken die Thevenin ekwivalent met verwysing tot terminale *a*-*b*. regh	
25	$\bar{V}_{TH}$ **(polar form)**	**25**	$\bar{V}_{TH}$ **(polar vorm)**
26	Z_{TH} **(rectangular form)**	**26**	Z_{TH} **(reghoekige vorm)**



$\underline{\bar{V}_{TH}}$

$$\frac{V_x - 2}{-j4} + \frac{V_x}{j12} + \frac{V_x}{j4} = 0$$

$$\cancel{\frac{jV_x}{4}} - \frac{j2}{4} - \frac{jV_x}{12} - \cancel{\frac{jV_x}{4}} = 0 \quad \rightarrow \text{Note: } \frac{1}{j} = -j$$

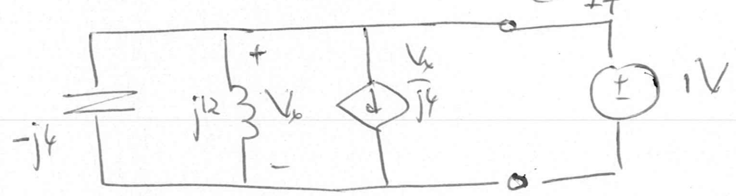
$$\frac{j}{2} = -j\frac{V_x}{12}$$

$$V_x = \underline{-6\ V} = V_{TH}$$

$$= \underline{6\angle 180°\ V}$$

$\underline{Z_{TH}}$

$\leftarrow I_T$ (Test source)



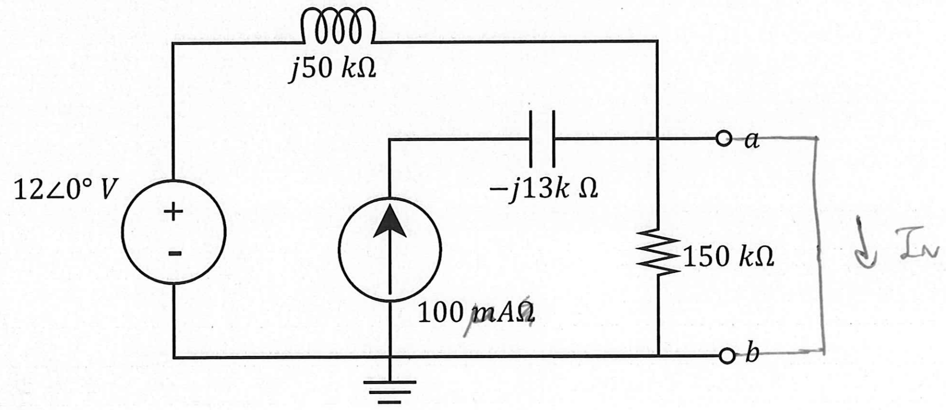
$* \ V_x = 1$

$$\rightarrow I_T = \cancel{\frac{V_x}{j4}} + \frac{V_x}{j12} + \cancel{\frac{V_x}{-j4}}$$

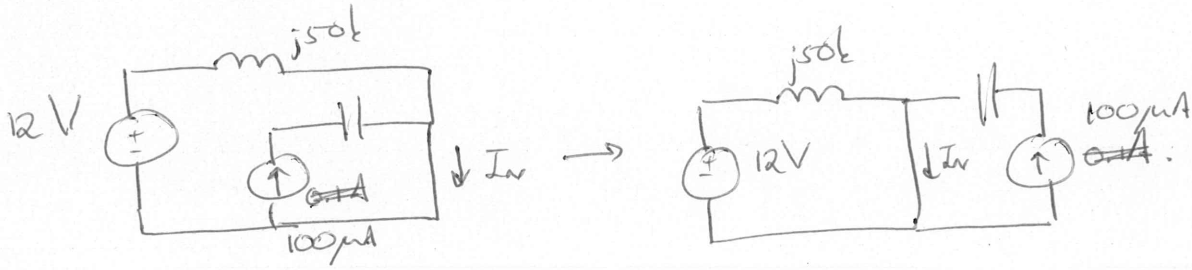
$$= \frac{V_x}{j12} = \frac{1}{j12}$$

$$\therefore Z_{TH} = \underline{j12\ \Omega}$$

Question 27 & 28 [4]		Vraag 27 & 28 [4]	
Calculate the Norton equivalent with respect to terminals *a-b*.		Bereken die Norton ekwivalent met verwysing tot terminale *a-b*.	
27	$\bar{I}_N$ (~~polar~~ form) rectangular	27	$\bar{I}_N$ (~~polêre~~ vorm) reghoekig
28	Z_N (rectangular form)	28	Z_N (reghoekige vorm)



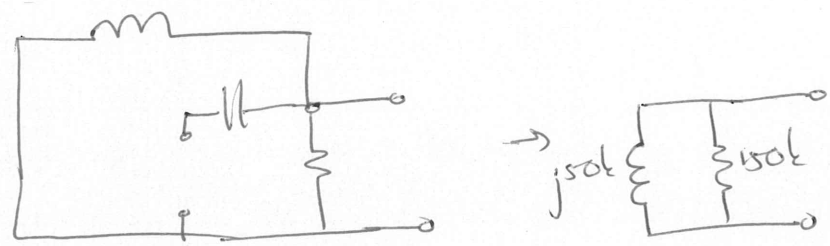
$\underline{\underline{I_N}}$ * Short circuit means 150kΩ can be removed since 0Ω||150kΩ = 0Ω.



$$I_N = 0.1\text{mA} + \frac{12}{j50k}$$

$$= \underline{100 - j240\ \mu A.}$$

$\underline{\underline{Z_N}}$



$$Z_N = (j50 \| 150)k = \left(\frac{150 \times j50}{150 + j50}\right)k = (15 + j45)k\Omega$$

PRACTICE PAGE / OEFENBLAD 1

Assessment ID	2	0	1	7	E	B	N	1	2	2	S	3		

	Answer		Unit
01	C_{ab} = 90		μF
02	$v_C(t = 0.1s)$ = −18.097		V
03	$v_L(t = 0.1s)$ = 9.516		V
04	w_C = 450		mJ
05	w_L = 30		mJ
06	*Statement #:* 2		*No unit required*
07	θ = −160		°
08	z = $1\angle -75°$		*No unit required*
09	$\bar{V}_T$ = $3\angle -45°$	**(polar)**	V.
10	$\bar{I}_T$ = $1.5\angle -45°$	**(polar)**	mA
11	$\bar{I}_0$ = 6 + j4	**(rectangular)**	A
12	$\bar{V}_0$ = −6	**(rectangular)**	V
13	ω = 2.5		krad/s
14	Z_{ab} = 6 + j10	**(rectangular)**	Ω
15	a = 6 − j5	**(rectangular)**	*No units required.*
16	b = j5	**(rectangular)**	
17	c = j4	**(rectangular)**	
18	d = 1 − j4	**(rectangular)**	
19	e = 10 − j4	**(rectangular)**	
20	f = ~~3~~ 5	**(rectangular)**	
21	g = 3	**(rectangular)**	
22	h = −1	**(rectangular)**	

PRACTICE PAGE / OEFENBLAD 2

Assessment ID	2	0	1	7	E	B	N	1	2	2	S	3		

	Answer		Unit
23	$k =$ j2	**(rectangular)**	*No unit required*
24	$Z =$ 1.2 − j3.6	**(rectangular)**	Ω
25	$\bar{V}_{TH} =$ −6	**(~~polar~~)** rectangular	V
26	$Z_{TH} =$ j12	**(rectangular)**	Ω
27	$\bar{I}_N =$ 100 − j240	**(~~polar~~)** rectangular	μA
28	$Z_N =$ 15 + j45	**(rectangular)**	kΩ